Heat, cold and first hospitalisation after cardiovascular diagnosis in Hong Kong

An exploratory monthly study, 2013–2023

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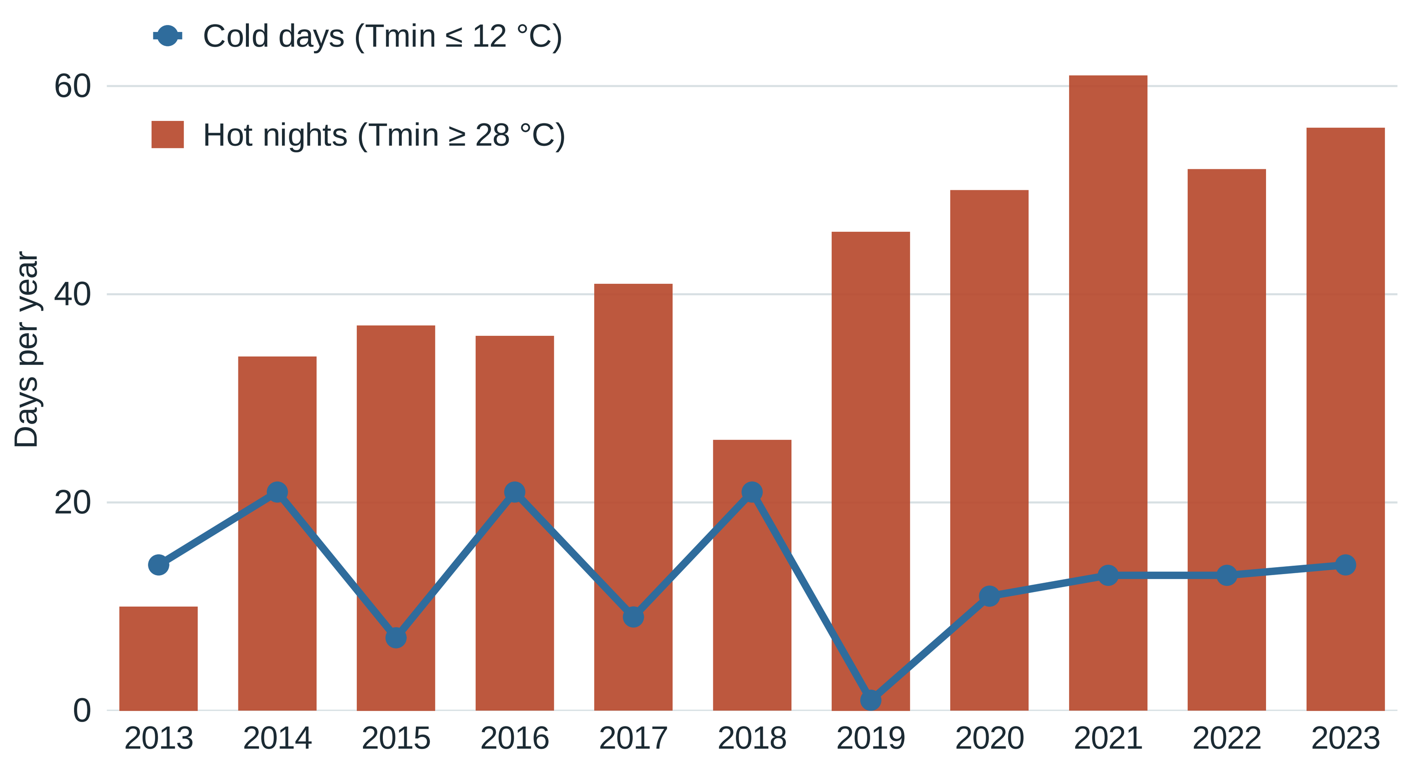
INTRODUCTION

Hong Kong's hot nights became more frequent over the past decade: at the Hong Kong Observatory they rose from 10 days in 2013 to a peak of 61 in 2021, while winter cold days have not disappeared. For the growing number of people living with type 2 diabetes or hypertension — a group at high cardiovascular risk — hospital burden complements the better-studied mortality burden. This study examines the monthly association between temperature, extreme-day burdens and first hospitalisation after cardiovascular diagnosis in that group.

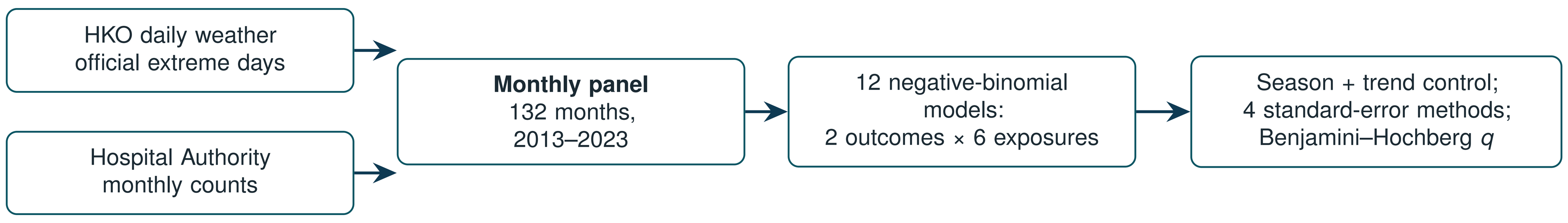


Official Hong Kong Observatory extremes at Headquarters. Hot nights rose about fivefold over the decade; cold days recurred every winter — the exposure setting for this study.

OBJECTIVE

Are monthly temperature and extreme-day burdens associated with **first hospitalisation after a recorded coronary heart disease (CHD) or heart failure (HF) diagnosis** in a high-risk Hong Kong cohort?

METHODS



- **Window and weather:** 132 territory-months, Jan 2013 – Dec 2023; daily records and official extreme-day definitions from the Hong Kong Observatory.
- **Outcomes:** privacy-protected Hospital Authority monthly aggregate counts of *first hospitalisation after a first recorded CHD or HF diagnosis* in people with type 2 diabetes and/or hypertension — 156,156 CHD and 29,681 HF events. Delivered columns are labelled inpatient; admission cause is not recorded, and final event semantics require data-provider confirmation.
- **Models:** 2 outcomes × 6 exposures = 12 negative-binomial regressions with calendar-month indicators, a trend spline, and a days-in-month offset; the panel was specified after the outcome series was available.
- **Uncertainty:** four standard-error methods per contrast; Benjamini–Hochberg q across all 12 comparisons.

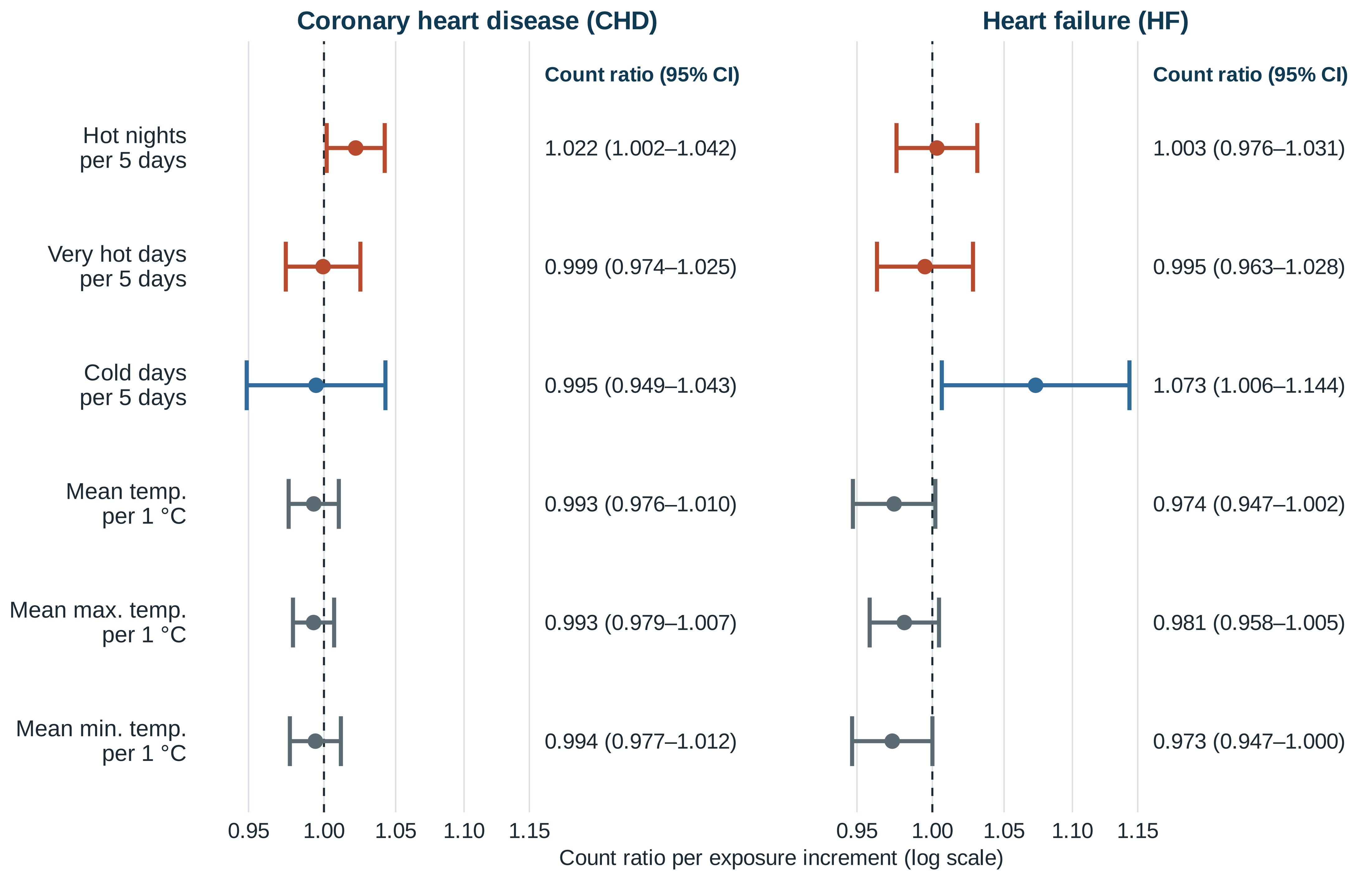
Definitions. Hot night: minimum 28 ℃ or higher; very hot day: maximum 33 ℃ or higher; cold day: minimum 12 ℃ or lower (HKO definitions). Estimates are count ratios per 5 extreme days or per 1 ℃.

RESULTS

An exploratory panel of twelve models, specified after the outcome series was available — two contrasts stand out.

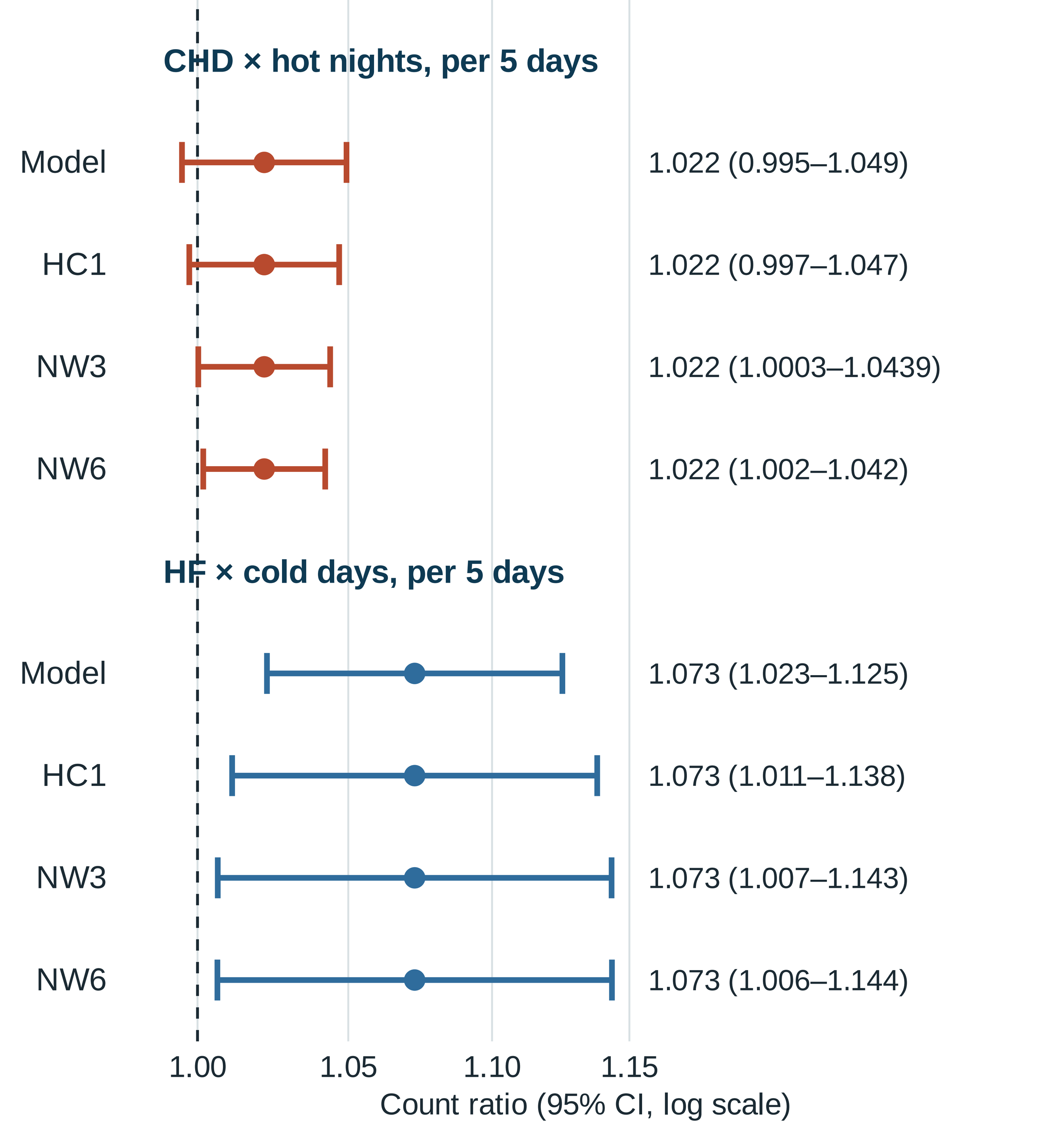
1.022	**CHD series × hot nights,** per 5 days 95% CI 1.002–1.042 $p = 0.032$ $q = 0.192$
1.073	**HF series × cold days,** per 5 days 95% CI 1.006–1.144 $p = 0.031$ $q = 0.192$

- **CHD series × hot nights:** about **2.2% higher** monthly first-hospitalisation count per five additional hot nights — but the interval *includes 1* under two of four standard-error methods, so the signal is uncertainty-sensitive.
- **HF series × cold days:** about **7.3% higher** monthly first-hospitalisation count per five additional cold days — the interval excludes 1 under all four methods.
- The other ten contrasts sit near 1. **All 12 Benjamini–Hochberg *q*-values exceed 0.19**: after correction for 12 comparisons, both signals remain exploratory.



The twelve-model exploratory panel: count ratios from single-exposure negative-binomial models (days-in-month offset); 95% intervals with Newey–West lag-6 standard errors. Values right of 1 mean more first hospitalisations following recorded diagnosis in months with higher exposure. All 12 Benjamini–Hochberg $q > 0.19$.

How robust are the two signals?



The two leading contrasts under four standard-error methods. For CHD × hot nights: Model and HC1 intervals include 1; NW3 excludes it only on the unrounded scale (1.0003–1.0439); NW6 excludes it. The HF cold-day interval excludes 1 under all four. The main display uses NW6; methods were not chosen by interval position.

INTERPRETATION / LIMITATIONS

- All 12 q-values exceed 0.19 — these are **exploratory signals**, not confirmed effects.
- The CHD hot-night estimate is **sensitive to how uncertainty is computed**; the HF cold-day estimate is stable across methods.
- Monthly, territory-level data with no recorded admission cause support **no causal or principal-diagnosis claim**.

CONCLUSION

Monthly data suggest **different thermal patterns** in the CHD and HF series — hot nights for CHD, cold days for HF — but the evidence is **exploratory, not confirmatory**: no contrast survived correction for twelve comparisons. The project leaves a **reproducible monthly framework** and a clear next requirement: **daily or cohort data with recorded admission cause**.

What would strengthen this: linked daily weather and admission records with cause of admission, so the monthly burden patterns seen here can be tested at the resolution at which heat and cold act.

Laidlaw Stage 3 Research Poster · ISO A0 portrait · Weather: Hong Kong Observatory daily records (public) · Outcomes: privacy-protected Hospital Authority monthly aggregate counts
Collaborators: Hogan (weather definitions) · Zhenyuan Liu / Roro (Hospital Authority aggregates) · Supervisor: Prof. David Bishai · Reproducible, machine-validated analysis and release package

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